

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

07

Q1

ALGEBRA

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy -plane. What is the slope of line j ?

- A. $\frac{7}{17}$
- B. $\frac{17}{7}$
- C. 4
- D. 17

A candle is made of 17 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 6 ounces of wax remain in this candle, for how many hours has it been burning?

- A. 3
- B. 6
- C. 24
- D. 44

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

- A. 0.05
- B. 12.1
- C. 20
- D. 242.0

A bus traveled on the highway and on local roads to complete a trip of 160 miles. The trip took 4 hours. The bus traveled at an average speed of 55 miles per hour mph on the highway and an average speed of 25 mph on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

- A. $55x + 25y = 4$
 $x + y = 160$
- B. $55x + 25y = 160$
 $x + y = 4$
- C. $25x + 55y = 4$
 $x + y = 160$
- D. $25x + 55y = 160$
 $x + y = 4$

A number x is at most 17 less than 5 times the value of y . If the value of y is 3, what is the greatest possible value of x ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

When line n is graphed in the xy -plane, it has an x -intercept of $-4, 0$ and a y -intercept of $0, \frac{86}{3}$. What is the slope of line n ?

A. $\frac{3}{344}$

B. $\frac{6}{43}$

C. $\frac{43}{6}$

D. $\frac{344}{3}$

$$a(3-x) - b = -1 - 2x$$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?

- A. $a = 2$ and $b = 1$
- B. $a = 2$ and $b = 7$
- C. $a = -2$ and $b = 5$
- D. $a = -2$ and $b = -5$

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

A. -4

B. 8

C. 42

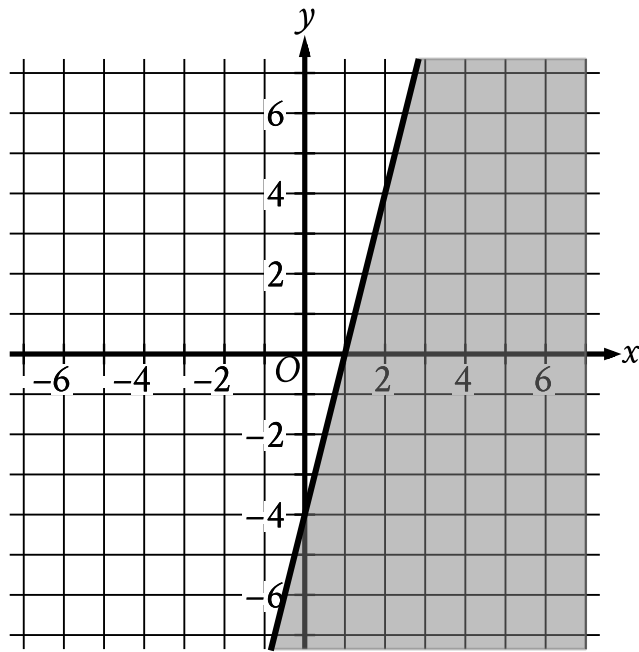
D. 44

$$4x + 5y = 100$$

$$5x + 4y = 62$$

If the system of equations above has solution (x, y) , what is the value of $x + y$?

- A. 0
- B. 9
- C. 18
- D. 38



- The boundary of the inequality is a solid line.
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (0 comma negative 4)
 - (1 comma 0)
 - (2 comma 4)
- The area below and to the right of the boundary is shaded.

The shaded region shown represents the solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

- A. -5, -6
- B. -2, 5
- C. 1, 4

D. 6, -2